Open PhD position in cognitive neuroengineering: decoding visual attention with high-resolution fMRI and artificial neural networks

The Laboratory of Synthetic Neuroimaging (Prof. A. Example: https://example.org/lab) is opening a new PhD position on the neural mechanisms of visual attention, studied with artificial neural networks and high-resolution fMRI recordings of the human visual cortex.

Project description:

The project combines model-based analyses of fMRI signals with behavioural experiments in healthy volunteers and patients with attentional deficits. The candidate will build encoding models that predict voxel-wise responses from image statistics and will test whether the same models account for attention-driven gain changes in early visual areas. Data collection takes place at the institute's imaging centre, and the analyses are developed in an open-source software stack.

This position is embedded in a larger research programme on attention and consciousness, and benefits from shared computing infrastructure and weekly methods seminars.

Requirements:

Applicants should hold a Master degree in neuroscience, computer science, physics or a related field, have strong quantitative skills (Python, PyTorch), and be comfortable with scientific writing. Prior work with fMRI preprocessing pipelines is an advantage but not a requirement. Please see https://example.org/how-to-apply for the formal procedure.

The position is funded for four years and includes a travel budget for international conferences. Part-time arrangements can be discussed.

Working environment:

The successful applicant will join a small interdisciplinary team with access to a 3T scanner, a high-performance computing cluster and a dedicated behavioural testing lab. The working language is English, and supervision is shared between two senior researchers. Inquiries may be sent to contact@example.edu.

Start date:

Fall 2027

Application procedure:

Interested candidates submit their application through the graduate school portal (https://example.org/graduate-school) and name two referees. Applications are reviewed on a rolling basis until the position is filled.